

Highlights

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

Yunus Emre Cogurcu, Emrehan Metin

- Pre-fire thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- The same block adds +0.004 AUC across 20 transfer directions, interval spans zero
- Highest niche-overlap pair fails both ways; lowest transfers both ways (points)
- Of 20 diagnostics, only conditional direction agreement tracks transfer (rho 0.84)
- Label-blind adaptation compresses transfer toward chance; only target labels help

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

Yunus Emre Cogurcu^{a,*}, Emrehan Metin^a

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness separates a fire year from a normal year. Models built on it are rarely tested outside the region they were fitted in. Five Mediterranean wildfire regions were analysed here, on about 500 m cells, with MCD64A1 labels and spatially blocked validation. Six pre-fire thermal predictors were added to a terrain, fuel and greenness baseline.

The local skill does not travel, and almost all of it is lost before the region changes. Within-region ROC-AUC rose by +0.056 to +0.153 in every region. Across twenty ordered transfer directions the same block contributed +0.004 [−0.028, +0.036]. That is not distinguishable from zero, and its sign changes from pair to pair. The static baseline transferred no better, at 0.537 against 0.541. Three evaluations on **identical cells** locate the failure. A model that has the held-out burn scar in its training data scores 0.627; withholding that scar gives 0.552; and a model fitted 306 to 2,802 km away gives 0.559. The fire-specific residual is +0.082 [+0.005, +0.159] and the effect of the 2,800 km is **−0.003 [−0.063, +0.056]**. The failure is therefore

*Corresponding author.

Email address: ycogurcu@cu.edu.tr (Yunus Emre Cogurcu)

a property of contiguous spatial holdout, not of separation distance and not of crossing a region boundary. Two thirds of the apparent collapse from a region-wide 0.782 is the evaluation area itself, which is 34 to 87 % burned with every negative fire-adjacent. Two predictors reverse their association between regions: elevation and the LST anomaly. Removing them cost -0.081 of within-region skill, supported in every region and mostly due to elevation. It changed transfer by $+0.014$ $[-0.017, +0.045]$. Both figures are post-selection estimates. A local cost is measured. No compensating transfer gain is.

The failure is invisible to the diagnostics that can be run before deployment. Twenty candidate diagnostics were rank-correlated against observed transfer. Only two had intervals excluding zero, and both were conditional. The stronger one measures agreement in the sign of each predictor’s association (Spearman $\rho = 0.84$ over sixteen directions). No marginal measure was shown to order the matrix, including predictor-space dissimilarity. At the point estimates, the pair with the highest burned-niche overlap failed in both directions, while the lowest transferred in both. Signed associations need labels on both sides. The family that works is therefore the one a practitioner does not have.

The mechanism is a reversal of sign. Predictors do not weaken across regions. They reverse the direction of their association with burning. A distance in predictor space cannot see that. The same reversal appears inside one study area, between two fires eleven months apart on an identical grid. That arm rests on one fire and eleven positive-carrying blocks, so it corroborates rather than establishes.

Label-free alignment pushed fourteen of twenty directions towards chance

rather than repairing them. Transfer skill has to be measured, not inferred from similarity. The price is target labels: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target’s matched ceiling in three of six directions. *Keywords:* wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [27]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [39, 13]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [4]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also, as we show, not portable: the relationship they carry is reparameterised locally, existing everywhere but pointing in different directions in different places. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair

20 rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own earlier framing took it. The data do not support it. The static baseline transfers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing
25 predictors returns +0.014 of transfer on an interval that also spans zero. Two nulls on the portability axis are not an exchange. What is measured is a local gain that does not export, and a local cost when the reversing predictors are removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

30 A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [31, 29] and is addressed
35 by spatially blocked cross-validation [37, 23]. Blocking removes the inflation that random folds produce, and it is the right correction for what it corrects. It does not make the estimate honest about a fire the model has not seen. Section 4.3 shows the size of that gap: blocked cross-validation at 5 km returns a mean of 0.797, and the same models applied to a burn scar held
40 out of their own region return 0.552.

Because only that side of the ledger is reported, portability is never entered at all. A predictor block is adopted on the strength of the increment it delivers inside its training footprint, and whether that increment survives a change of region is not asked, even though any regional product built

45 from locally trained models implicitly promises generalisation beyond that
 footprint. For fire specifically the transfer question is asked far less often
 than in adjacent fields. Meteorological fire-danger indices are known not
 to port cleanly between fire environments [30], and the two recent studies
 that test model transfer systematically, across Mediterranean countries [7]
 50 and US counties [15], both report that it largely succeeds between similar
 regions. Both transfer models whose dominant predictors are *spatially sta-
 tionary*: terrain, human modification, night-time lights, population density,
 long-term climatologies. Each describes a place rather than a season, which is
 the class our thesis expects to travel well. Species distribution modelling has
 55 examined transferability far more systematically [43], and there geographic
 and environmental similarity do *not* reliably predict transfer success [38, 32].

1.2. Pre-fire thermal dryness is the natural test case

The dynamic class used here is pre-fire thermal dryness, and it is the
 class most plausibly *expected* to transfer. Standard susceptibility predictors
 60 are terrain, fuel type, greenness, long-term climatology and proximity to
 roads and settlements, and most are static or near-static over the timescale
 at which fire danger varies. Such a set explains poorly why a particular sum-
 mer burned and the preceding one did not. What changes between those
 summers is the state of the surface: fuel dryness, moisture stress, and how
 65 anomalously hot the land surface has become relative to its own baseline.
 Satellite thermal observation gives direct access to part of that state through
 fuel moisture content [44], and pairing surface temperature with a vegeta-
 tion index was established as a live-fuel-moisture estimator for fire-danger
 rating by Chuvieco et al. [6]. The Temperature-Vegetation Dryness Index

70 [33] formalises that feature space into an internally normalised measure. In principle it should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. That theoretical portability advantage is tested here, and it is not found: the two internally normalised channels transfer no better than the four absolute ones, at means of 0.544
75 and 0.548 over the twenty directions (Appendix A(f)).

That pre-fire thermal state carries genuine information about subsequent fire is established [17, 19, 18, 8]. What has not been established is whether the predictive *skill* of a classifier trained on it in one fire region survives strict, label-free application to another. The physics linking moisture stress
80 to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate.

1.3. Why the loss is invisible to the diagnostics in use

A deficit on transfer can arise two ways. Under **covariate shift** the
85 marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [35], an approach with an established remote-sensing literature [36]. Under **concept shift** the conditional relationship itself changes: a predictor pos-
90 itively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [25].

This distinction is what makes the failure hard to anticipate. The di-

95 agnostics the field relies on to decide whether a model may be applied to
new ground, above all area-of-applicability measures built on dissimilarity in
predictor space [21, 22, 16], are marginal by construction. A target region
can sit well inside the training data’s predictor envelope while the relation-
ship between those predictors and burning points the other way. This paper
100 measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval
supports, and the nearest prior work is named inside the claim.

Contribution 1. Local skill and portability, measured separately
105 **and per direction.** The thermal block is worth +0.056 to +0.153 ROC-
AUC inside every one of five regions, with every bootstrap interval above
zero and the result holding in both analysis populations. Across the twenty
ordered transfer directions it contributes +0.004 $[-0.028, +0.036]$. That
interval spans zero under all four between-direction resampling units we
110 computed, and the sign varies by pair: paired deltas run from -0.148 to
 $+0.132$, twelve positive and eight negative. Two controls bound the reading.
The baseline arm transfers at a mean of 0.537, against the thermal model’s
0.541, so the static predictor class is not the portable one either. Three
evaluations scored on **identical cells** locate where the skill goes. A model
115 with the held-out burn scar in its training data returns 0.627 on that scar’s
area; withholding the scar returns 0.552; and a model fitted 306 to 2,802
km away returns 0.559. The fire-specific residual is +0.082 $[+0.005, +0.159]$;
the effect of the 2,800 km is **-0.003 $[-0.063, +0.056]$** . **The failure is a**
property of contiguous spatial holdout, not of separation distance and

120 not of crossing a region boundary. It is not shown to be a property of the
 fire event either, because the held-out patch is defined by the labels and its
 identity cannot be separated from its location. Two thirds of the apparent
 fall from the region-wide 0.782 is the evaluation area, which is 34 to 87 %
 burned against 3.8 to 28.7 % for a region, so a scar-level result compared
 125 against a region-level reference overstates the collapse. Separation does not
 order the matrix either, at $\rho = -0.32$ with an interval spanning zero. Six
 directions are nonetheless below chance with interval support, which no ac-
 count of merely lost skill explains, and that residual is what Contributions
 2 and 3 address. Feature removal measures the local cost. Dropping the
 130 two reversing predictors, elevation and the LST anomaly, costs -0.081 of
 mean within-region AUC, supported in every region. Roughly three quarters
 of that is elevation, a *baseline* terrain variable. Mean transfer changes by
 $+0.014$, whose interval also spans zero. Both are post-selection estimates
 (Section 4.6b). A local cost is measured. No compensating transfer gain
 135 is. The within-region half is not novel, comparable results existing for these
 landscapes [1, 12]; the paired contrast against portability is. Dimarco et al.
 [7] transfer a predominantly stationary predictor set successfully across a
 comparable Mediterranean design, though their response variable is ignition
 rather than burned area (Section 5.7).

140 **Contribution 2. The loss is invisible to the diagnostics that
 could be run before deployment, and visible to one that cannot.**
 Twenty candidate diagnostics from four families are evaluated against ob-
 served transfer under one bootstrap framework. Area-of-applicability-style
 predictor-space dissimilarity [21, 22, 16], climatic and geographic distance,

145 learned domain separability, niche-overlap statistics and burn-pattern regime
 distances were none of them shown to order the matrix; the domain classifier
 is at ceiling, with $AUC \geq 0.96$ for every pair. Only two diagnostics have in-
 tervals excluding zero. Both measure conditional direction agreement, that
 is, whether each predictor’s signed association with burning points the same
 150 way in both regions (Spearman $\rho = +0.84$ and $+0.81$ over sixteen directions
 from eight pairs). Both are computed on a data-selected subset of predic-
 tors, and their all-nine-feature counterparts span zero, so the result rests
 on that selection. Equalising the families onto a common twelve directions
 leaves the ordering unchanged, so it is not an artefact of unequal samples.
 155 Signed associations need burned labels on both sides, so this is a mechanism
 diagnosis rather than a pre-deployment screen, and the marginal family, the
 only one runnable before deployment, is the one that fails. The same point
 survives without any ranking, at the point estimates: the pair with the high-
 est burned-niche overlap fails in both directions while the lowest transfers
 160 in both. That contradicts an expectation the fire literature carries [7, 15]
 and agrees with species distribution modelling [38, 32]. Shift decomposition
 in applied remote sensing is not itself new [11]; the fire application and the
 head-to-head of four diagnostic families on the same pairs are.

Contribution 3. The mechanism is a reversal of sign, and it sur-
 165 **vives holding geography fixed.** Predictors do not merely weaken across
 regions, they reverse the direction of their association with burning, which is
 why a distance in predictor space cannot see the failure. The sharpest rever-
 sal is elevation, whose association points opposite ways in Manavgat and in
 Bejís and Muğla, with each region’s own interval excluding chance and the

170 paired difference excluding zero (Appendix B, Table B3). The same reversal
appears inside a single study area, between two fires eleven months apart
on an identical grid, where season and population also differ but place does
not (Section 4.8). That arm rests on one fire and eleven positive-carrying 5
km blocks, below the floor this design sets for itself, so it corroborates the
175 mechanism rather than establishing it.

Two consequences follow. Both are reported as supporting results, not
leading ones. First, label-free alignment by standardisation and covariance
alignment [35] does not repair transfer. It compresses transfer towards chance
in fourteen of twenty directions. We believe this is the first application
180 of covariance alignment to fire susceptibility. Second, the residual gap is
conditional, so the resource that closes it is target labels. That price is
measurable. Thirty-two labelled 5 km blocks recover 85 to 89 % of the
target’s matched ceiling in three of six directions, 51 to 57 % in two more,
and 30 % in the sixth.

185 Alongside these, a leakage-audited, spatially blocked evaluation and trans-
fer protocol is released with code, configuration and frozen outputs, so that
most of this result can be re-run rather than taken on trust. One component
is an exception and is named in the data-and-code availability statement:
the reproduction-check code is not in the released repository at the commit
190 of record. The few-shot export, previously also listed as an exception, was
regenerated at commit 6d7a6a71 and its validator now passes on every check.
This matters given evidence that wildfire transfer conclusions are sensitive
to evaluation design and task formulation [42].

A companion paper treats the observational layer beneath this one, where

195 preprocessing decisions taken before any model is fitted are shown to carry
their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study
regions, the data, and the modelling, cross-validation, transfer and adapta-
tion protocol. Section 4 reports results. Section 5 discusses them, including
200 limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both method-
ological and application perspectives [39, 13]. The dominant pattern is sta-
205 ble: geospatial predictors are assembled over a study region, paired with a
historical burned-area record, and fitted with a supervised classifier, most
often a random forest [4], and the resulting surface is published with a cross-
validated AUC in the 0.85 to 0.95 range. That figure is a within-region
estimate. Comparable within-region results already exist for the landscapes
210 studied here [1, 12].

2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through
fuel moisture content [44]. Pairing land surface temperature with a vegeta-
tion index was established as a live-fuel-moisture estimator for fire-danger
215 rating by Chuvieco et al. [6], and the Temperature-Vegetation Dryness In-
dex [33] formalises that feature space into an internally normalised measure,
which is why it is often expected to travel between regions better than raw

temperature. That pre-fire thermal state carries information about subsequent fire is established [17, 19, 18]. Gelabert et al. [8] found dead fine fuel
220 moisture and its anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

225 2.3. *Spatial validation and transferability*

Random cross-validation over spatially autocorrelated cells inflates skill estimates [31, 29], and spatially blocked designs are the standard remedy [37, 23]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [40, 24, 5]. This paper uses spatial
230 blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking corrects the inflation that random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures and finds to be a larger gap than the blocking correction itself. For that, the field’s instrument is the area of applicability and
235 related predictor-space dissimilarity measures [21, 22, 16], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [43], and two studies there report that geographic and environmental
240 similarity do not reliably predict transfer success [38, 32], and the transferability of satellite-derived empirical models is an active question in fire remote sensing itself [20].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor
distribution moves but the predictor-response relationship holds, from con-
cept shift, where the relationship itself changes [25]. Covariate shift is in
principle correctable without target labels, by per-region standardisation or
by covariance alignment such as CORAL [35], and domain adaptation has
an established remote-sensing literature [36, 28]. Concept shift is not: no
realignment of input distributions can repair a reversal in the sign of an
association, because detecting it requires the labels being withheld. Shift
decomposition in applied remote sensing is not itself new [11]. We are not
aware of a prior application of covariance alignment to fire susceptibility, fire
occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [30] re-
port, in a Peruvian case study, that meteorologically derived danger indices
do not port cleanly between fire environments. WildfireGenome [15] runs
a leave-one-county-out matrix across seven US counties and reports strong
within-county performance with highly variable off-diagonal transfer, similar
pairs transferring well and dissimilar pairs collapsing; its label, however, is
a principal-component composite of hazard *indicators* rather than observed
burned area, and it applies no adaptation. Xu et al. [42] argue that wild-
fire transfer conclusions depend strongly on evaluation design, a caution we
address by fixing the protocol in a project log before the diagnostics were
computed and by reporting every sensitivity axis; that log is not a formal pre-
registration, and Section 3.11 states what was fixed and when. Kondylatos

et al. [14] provide Mesogeos, a 1 km Mediterranean datacube for data-driven wildfire modelling.

270 **The nearest neighbour, and the contrast the present paper draws.**

Dimarco et al. [7] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar
275 countries scoring higher. They apply no domain adaptation.¹

Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state
280 of a surface in one season. **The response variable:** theirs is human-driven ignition, ours is burned area, and those are different quantities with different dominant controls. Read together the two results suggest that the relationship between domain similarity and transfer success may be predictor-class dependent, with Vesk et al. [38] and Rousseau and Betts [32] aligning from
285 species distribution modelling while Dimarco et al. and WildfireGenome run against it. We present that as a live disagreement, and Section 5.7 states why our own data do not settle it.

Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology [26]. We follow their Methods, which we take to be authoritative, and
290

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

note the discrepancy so that the comparison is transparent. Every statement about their work was checked against the full text, read 2026-08-08.

3. Data and methods

3.1. Study regions and temporal windows

295 Five Mediterranean wildfire regions are analysed (Fig. C.1): Manavgat 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any
300 model metric. One AOI choice was label-informed and is stated as such. The North Evia box was extended after the legacy box was found to carry an atypically high burned prevalence. The extended geometry was then defined from place anchors, and the legacy variant is kept as a sensitivity arm (Appendix A). The AOIs are deliberately not clipped to fire perimeters,
305 so that unburned cells around each fire form the negative class rather than being excluded by construction. A sixth region, Kozan 2023, is carried as a negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation
310 can post-date the first labelled burning. Window lengths vary between regions with the event: 57 to 61 days for predictors and 35 to 59 days for labels, counted inclusively. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and

315 baseline years are registered in `core/regions.py` and are reproduced here.

Table 1: Study regions, areas of interest and temporal windows. Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (56 d)	2021-07-28 to 2021-08-31 (34 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (60 d)	2022-08-15 to 2022-09-30 (46 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (57 d)	2021-07-29 to 2021-09-15 (48 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (58 d)	2021-08-03 to 2021-09-30 (58 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (59 d)	2021-07-24 to 2021-08-31 (38 d)	2017, 2018, 2019, 2020

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 [9] retrieved through Google Earth Engine [10], whose omission and commission characteristics [3] bound every model here. The analysis grid is **reconstructed rather**
320 **than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s
325 eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran

for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

330 3.3. *Burned-landcover admissibility gate*

Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. 335 The purpose is to separate burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. *Predictor variables*

Ten predictors are used, four baseline and six thermal, all summarised 340 per cell over the predictor window, through the chain shown in Fig. C.2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/O61/MOD11A1.

- 345 • **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 [45], entering as the dominant class code.

- 350 • **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- 355 • **TVDI and TVDI difference**. The Temperature-Vegetation Dryness Index [33] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference` = `current_tvdi` - `mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.
- 360
- 365
- **Downscaled LST and fused LST**. A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler’s own inputs
- 370

include coordinates, which is the one route by which a coordinate-
375 derived surface re-enters a feature set from which Section 3.14 excludes
coordinates. Appendix A(g) reports the increment without these two
channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid
380 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_`
`modeling` when it meets a 30 % valid-pixel floor and no thermal channel
is missing. Two populations are then defined. The **primary** population
is natural vegetation, cells whose combined tree, shrub and grass fraction
reaches 0.50, which excludes cropland from every burnable mask. The **sec-**
385 **ondary** population is all valid cells, reported throughout as a sensitivity.
Every headline result is given for both.

3.6. Feature sets and classifier

Two nested feature sets are compared. The **baseline** set is NDVI, el-
elevation, slope and land cover. The **thermal** set is the baseline plus the
390 six thermal channels. Land cover is one-hot encoded and numeric features
are standardised inside the fitting pipeline. The classifier is a random for-
est [4] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced
class weights and `random_state = 42`, identical for every region, popula-
tion, feature set and transfer direction, so that no comparison in this paper
395 is confounded by a model choice.

3.7. *Spatial-block cross-validation and bootstrap uncertainty*

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are
400 used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's
405 reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

3.8. *Cross-region transfer protocol*

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used**
410 **at any point**: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal block's contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five
415 regions are computed.

3.9. *Label-blind domain adaptation*

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region's numeric features are standardised using its own statistics, source statistics from source data and target statistics

420 from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [35], with regularisation $\lambda = 10^{-5}$ and eigen-
 425 values floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.

430 λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies outside the released sweep and is reported separately in Appendix A(b).

3.10. *Transfer-gap decomposition and the concept-shift diagnostic*

435 For each direction the gap between the target’s own within-region skill and the raw transfer result is split in two. One part is what the best label-free adaptation recovers, and the other is what it does not. The recovered fraction is defined as $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, and its interval comes from the same paired bootstrap. The recovered part bounds
 440 what covariate-level correction can achieve. The remainder is the conditional residual.

The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a

445 value below 0.5 is read as a direction rather than as weakness. A reversal is called bootstrap-supported only when the two regions' intervals are disjoint, under the same 10-cell spatial-block bootstrap.

3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from four families are computed for every
450 region pair and rank-correlated (Spearman) against observed raw transfer AUC, under one bootstrap framework that resamples unordered region pairs with both of their ordered directions travelling together. The families are marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity, climatic and geographic distance and a learned domain classifier;
455 burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and conditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with the result. Appendix C.2 gives the full specification of each diagnostic and of the bootstrap.

460 3.12. *Interventions*

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

465 **Removal of direction-reversing features.** The two predictors whose signed association reverses between regions with bootstrap support are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean**

(Bejís against Evia). They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the
 470 cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then read, so both quantities are post-selection estimates and no correction for that selection is applied.

475 3.13. Controls on the transfer path

Three controls establish what the transfer arms are measuring. All three use the transfer protocol of Section 3.8 unchanged: fit on the source cells, apply to the target cells, no refit, no recalibration, no threshold selection. All three were verified to reproduce the frozen cross-region AUCs exactly before
 480 being used.

Within-region half-split. The modelled cells of a region are cut in two at the median of a grid axis, a model is fitted on one half and applied to the other. Both axes and both directions are run, giving four splits per region. A split is discarded when either half is single-class, which happens twice in
 485 one region. Per-split source and target positive counts are reported, because they are unequal and bear on the interpretation.

Distance stratification. Because the cut is a straight line, a target cell's distance to the nearest training cell equals its distance to that line, so one fitted model per split yields an AUC at every separation. Target cells
 490 are binned at 0, 5, 10, 20, 40, 80 and 160 km, a bin is used when it holds at least 50 cells of both classes, and results are reported both unweighted over bins and weighted by burned cells, because the two disagree.

Leave-one-scar-out. Burned connected components are found with 8-connectivity on the analysis grid. Every component of at least 50 cells is
495 held out together with all cells within a buffer of it, a model is fitted on the remainder of the same region, and applied to the held-out area. Buffers of 2, 5 and 10 km are run. This is the control that separates the fire event from separation distance, since a scar held out at 2 km inside its own region is close to its training data and yet wholly unseen.

500 Appendix A(h) reports all four in full, with per-split and per-scar tables.

3.14. *Leakage control*

An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all
505 feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses
510 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly, and the twenty transfer directions agree to within 1.6×10^{-7} . Random-
515 forest fits are not bit-identical across library versions. All numbers here were therefore produced under scikit-learn 1.9.0, or verified against it. If that

version is not fixed, cross-region point estimates carry an implementation tolerance of about ± 0.02 to 0.03 . The companion paper reports the version
520 sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices, the dependence is reported rather than resolved by choosing the favourable
525 setting.

3.15. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical analysis grid, eleven months after the first. Signed univariate AUCs are computed
530 for both arms under the same 10-cell spatial-block bootstrap used elsewhere. Season, year and population all differ between the arms, since the 2022 arm is defined by removing the 2021 scar, and Section 4.8 reports what that costs. Appendix C.3 gives the window dates, the population construction and the asymmetry audit.

535 **4. Results**

4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire candidates and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is ex-
540 cluded from all modelling. The separation is not marginal: the five admitted

regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, though one
545 negative control cannot establish it.

Table 2: Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejis 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Mugla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Patri 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to

0.287. The effect of that choice on transfer is reported in Appendix A(a).

550 *4.2. Within-region: the thermal increment replicates in five regions*

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region. The increment's bootstrap interval excludes zero in all five regions at 1 km and at 5 km blocking (Fig. C.3). Those are the two scales this design supports as intervals. At 10 km the
555 point estimates hold, from +0.048 to +0.154. They rest on 6 to 33 positive-carrying blocks, so they are reported as indicative rather than as intervals. The table note gives the reason.

Table note (resampling units). The bootstrap resamples spatial blocks, so what bounds an interval's reliability is the number of blocks that carry
560 at least one burned cell. Those counts fall quickly as blocks coarsen. At 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No
565 bootstrap replicate was invalid at any block size, with one exception. At 20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom of the six positive-carrying blocks just reported, and a further reason to read the row as indicative. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells
570 also feeds only 12 groups into a 5-fold grouped split, so its models train on about ten blocks each. **The 20-cell row of this table should be read as indicative rather than as an interval.** The 10-cell row is the coarsest blocking this design supports properly, because every region there has 16

Table 3: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

to 70 positive-carrying blocks. The increment holds at that scale in all five
575 regions. Source: `paper/referee2_numbers.md`, block C, counted from the
frozen per-cell prediction tables.

This within-region result is not itself novel. It is reported because the
transfer arms below are measured against it.

4.3. Cross-region transfer, and what label-free adaptation does to it

580 **Raw transfer is heterogeneous and includes anti-predictive di-
rections.** Raw target AUC spans 0.326 to 0.686 (Fig. C.4). Twelve of 20
directions are above chance with CI support. Six are *below* chance with CI
support: both directions of Manavgat to Bejís and of Manavgat to Muğla,
plus Bejís to Evia and Evia to Bejís. Two intervals span 0.5. Even the best
585 raw transfer, Evia to Manavgat at 0.686, stays far below that target’s own
within-region thermal performance of 0.870. Across all directions the raw
deficit against the within-region reference is 0.184 to 0.592 AUC.

Those counts belong to the 2-cell blocking of Table 6. At the more conser-
vative 10-cell blocking the same points give 9 above, 4 below and 7 uncertain,
590 and no direction changes side of the chance line (Section 4.7). Four of the
six below-chance directions keep their support there. Bejís to Manavgat and
Manavgat to Muğla lose it and carry no verdict. The qualitative statement
is unchanged. The counts should not be read as exact.

The static baseline does not transfer either. This is the control
595 for the reading the rest of the paper invites, and it constrains that reading
sharply. The same twenty directions were run with the terrain, fuel and
greenness baseline alone. The mean target AUC is **0.537**, against **0.541**
for the thermal model. Static attributes of a place are the class Dimarco

et al. transfer successfully, and are the class this paper’s framing treats as
600 portable. Here that class is itself barely above chance. The paired per-
direction contrast is given in Section 4.6b. What matters here is that the
transfer failure below is not specific to the dynamic block. A baseline that
does not travel, plus pre-fire thermal state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its
605 **interval.** The two matrices were differenced direction by direction. The
mean is **+0.004**. The directions are not independent, because each region
appears in eight of the twenty. The interval therefore depends on what is
treated as the resampling unit. All four units the design permits give the
same answer:

Table 4:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	[−0.027, +0.034]
Unordered pairs, cluster bootstrap	10	[−0.028, +0.036]
Unordered pairs, t on pair means	10	[−0.034, +0.042]
Regions, leave-one-out jackknife	5	[−0.037, +0.046]

610 Every interval spans zero, and the point estimate is a small fraction of
each width. The per-region jackknife shows how little the mean is anchored.
Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives
+0.0148, +0.0021, +0.0065, **−0.0081** and +0.0060. **Dropping Evia alone
reverses the sign of the headline.** None of these four units propagates
615 within-direction sampling variability. They resample between directions only.

Where the skill is lost, on a matched comparison. Four evaluations

are reported. The last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first is included to show why an unmatched comparison misleads. Each uses the transfer protocol unchanged:
620 fit, then apply with no refit, no recalibration and no threshold selection.

The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it. That area is 34 to 87 % burned, against 3.8 to 28.7 % for a whole region, and every negative in it is fire-adjacent, so it is a harder discrimination problem than a region.

Table 5:

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.782	[0.752, 0.811]
B. Same blocked model, restricted	the region, scar included	the scar area	0.627	[0.566, 0.688]
C. Leave-one-scar-out	the region, scar withheld	the scar area	0.552	[0.510, 0.593]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.559	[0.514, 0.604]

Most of the apparent collapse is the evaluation area, not the fire.
625 A and B use the same model and differ only in which cells it is scored on, and that alone accounts for 0.155 of the 0.230 fall from A to C, about two thirds. Any comparison of a scar-level result against a region-level reference inherits that, and the region-level reference is what a paper of this kind normally
630 reports.

Withholding the fire costs little, and moving 2,800 km costs

nothing measurable. B minus C, the fire-specific residual on identical cells, is **+0.082** [**+0.005**, **+0.159**]. C minus D, the effect of replacing a same-region model with one fitted between 306 and 2,802 km away, is **-0.003** [**-0.063**, **+0.056**]. The second is the cleanest result here: on the same cells, a model fitted in another country does as well as one fitted in the same region with the scar withheld.

What this establishes is that the failure is a property of **contiguous spatial holdout**. It is not a property of separation distance and not of crossing a region boundary. What it does not establish is the fire event as the unit, because the held-out patch is defined by the labels, so its identity cannot be separated from its location. Two further observations point the same way: a model trained on the other half of the same fire is no better at matched sample size, and the within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out's 0.552 with overlapping intervals.

Separation does not order the twenty-direction matrix either. The Spearman correlation between centroid separation and transfer AUC is -0.32 with an interval spanning zero, and the two nearest directions, Manavgat and Muğla at 306 km, are among the worst at 0.470 and 0.401. This is a different quantity from Table 8's geographic-distance row, computed on the twelve-direction subset, which gives -0.24 ; both span zero. The transfer mean of 0.541 is over all twenty directions and is not the value at any one separation: the 2,802 km pair returns 0.326 and 0.444.

The result is robust to how the held-out patch is defined, at 0.543 to 0.565 across minimum component sizes from 25 to 200 cells and

under both 4- and 8-connectivity, and flat across buffers of 2, 5 and 10 km.

The increment behaves the same way. Contribution 1 is about the paired thermal-minus-baseline difference. At the half-split separation it is
660 **+0.027** on average, positive in 13 of 18 splits and running from -0.255 to $+0.164$, between the within-region $+0.056$ to $+0.153$ and the cross-region $+0.004$.

Six directions are below chance with interval support, the sharpest at 0.326 [$0.305, 0.349$]. No account of merely lost skill produces a reliably
665 reversed ranking, so those six need a mechanism acting on the direction of the relationship. That is the subject of Sections 4.4 to 4.8. Per-split and per-scar detail is in Appendix A(h).

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions
670 span 0.431 to 0.630 and under CORAL 0.443 to 0.624 , roughly half the raw spread, with no adapted direction exceeding 0.631 against within-region references of 0.859 to 0.918 . Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began
675 and 6 end further from it; five of those six involve Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443 . The 14 to 6 split should be read at the precision of limitation (ix) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001 .

Against the right reference, adaptation is not failing (Fig. C.5).
680 The three controls above give an achievable reference for a model applied

Table 6: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

Table 7: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 6. The adapted arms were not recomputed at the coarser blocking of Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

to a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out. The best label-free adaptation averages **0.556** across the twenty directions, against 0.541 raw. It is therefore at that reference, not far below
685 it. What it does is regress the matrix towards that value: fourteen of twenty directions move closer to chance, which harms the directions that already worked and helps the ones that did not. The verdict that follows is narrower than "alignment fails". Alignment cannot exceed what a model can achieve on an unseen fire, and it does not.

690 In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-split control above shows that the larger part of that remainder is already incurred inside the region, so it should not be read as a measure of concept
695 shift. Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

4.4. *Transferability diagnostics: only conditional similarity orders transfer*

700 Twenty candidate diagnostics from four families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Only two diagnostics have intervals excluding zero, and both are conditional. The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84$ [+0.58, +0.88]. The cosine variant reaches
705 +0.81. No marginal measure was shown to order the matrix. That includes

Table 8: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) condi- tional	6	+	+0.84 [+0.58 , +0.88] agree- ment fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain- classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] ^{log} effective- N distance	no	no
geographic	1	−	−0.24 [−0.84,	yes	no

area-of-applicability-style dissimilarity in predictor space, climatic distance and geographic distance. The niche-overlap and regime families were not shown to order it either. The learned domain classifier is at ceiling, separating
710 source from target at $AUC \geq 0.96$ for every pair. It always succeeds, which is why it carries no ordering information.

Two limits are stated with the result rather than after it. The first is size. The index’s tie structure caps the achievable Spearman at +0.861, so the observed +0.840 sits essentially on that ceiling. Its exact one-sided
715 permutation p is 0.0060, the smallest this tie structure can produce, against a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared family-wise correction on ten effective pairs. The second limit is labels. Signed associations need burned labels in both regions. The family that works is therefore not available before
720 deployment, while the family that fails is.

A third limit is about selection, and it is the sharpest of the three. The two rows that clear zero are the *supported-feature* variants. Their feature subset is chosen by whether two regions’ bootstrap intervals happen to be disjoint. That is a data-dependent selection, made on the same data, with
725 no correction. The unselected counterparts over all nine features are $\rho = +0.50 [-0.17, +0.83]$ for the cosine and $\rho = +0.18 [-0.40, +0.72]$ for the agreement count. Both span zero. The result lives in the selection step, and is reported as such.

The families are compared on unequal samples, and equalis-
730 **ing them does not change the ordering.** The marginal, applicability, climatic and geographic rows sit on twelve directions, because those diag-

nostics exist only for the four-region subset. The supported-conditional rows sit on sixteen and the rest on twenty. A reader may reasonably ask whether the marginal family’s failure is a statement about power rather than about
735 diagnostics. Every row was therefore recomputed on the common twelve directions. The published values reproduce to 4.8×10^{-5} . The conditional rows still lead, at $+0.87$ $[+0.65, +0.88]$ and $+0.85$ $[+0.43, +0.88]$, and every marginal row still spans zero. The ordering is not an artefact of unequal samples.

740 4.5. *The contrast pair: similarity is not sufficient*

The clearest single view of Table 8 needs no ranking at all (Fig. C.8). Manavgat and Muğla lie in the same country and the same fire year. They are 307 km apart by the centroid geodesic distance this paper uses as a diagnostic, and their nearest boundaries are 191 km apart. Their burned cells
745 occupy the most similar environmental envelope of any pair in the matrix, with per-feature Schoener’s D of 0.77 to 0.89. Yet five of nine feature-response directions point opposite ways. Elevation is one of them, at a signed AUC of 0.374 $[0.289, 0.471]$ in Manavgat against 0.611 $[0.532, 0.690]$ in Muğla, with disjoint intervals. All four absolute thermal channels are among the others.
750 Transfer is below chance in both directions at the point estimate, at 0.470 and 0.401. Both directions also sit inside the nominal area of applicability, at 0.875 and 0.531 of target cells.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap
755 measure. They transfer above chance in both directions. That half of the contrast rests on point estimates: neither direction carries a verdict at 5

km blocking, and the marginal applicability audit was never produced for Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two
760 coexisting counterexamples, so it does not depend on the number of pairs available.

4.6. Interventions: pooling and feature removal

(a) **Pooled multi-region training** (Fig. C.6). At the point estimate, training on the pooled primary populations of the other four regions never
765 beats the best single-source transfer for any target, with shortfalls of 0.02 to 0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation does not recover what single-source transfer loses, and since it is evaluated
770 entirely outside the saturation radius its failure is what separation alone predicts.

(b) **Removing the direction-reversing features.** The two predictors whose signed association reverses between regions **with bootstrap support** are **elevation_mean** and **lst_anomaly_mean**. The per-region signed
775 associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining without them costs **−0.081** of mean within-region AUC, supported in every region (per-region deltas $-0.060, -0.130, -0.074, -0.063, -0.079$, every interval entirely below zero), and changes mean transfer by **+0.014** [**−0.017**, **+0.045**], which spans zero.

780 Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block’s. Dropping elevation alone accounts

for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone accounts for -0.013 , from 0.888 to 0.875. Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were chosen because they reverse, using the same data on which the -0.081 and the $+0.014$ are then estimated. No correction for that selection is applied.

Table 9:

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

Fig. C.7 shows the four configurations together. What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms. The thermal block’s own contribution to transfer is $+0.004$, with an interval spanning zero (Section 4.3). Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Two nulls on the portability axis are not a price paid.

4.7. Sensitivity analyses

Seven design choices were varied with everything else held fixed. They are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality screening of the coarse thermal input, the contrast between the normalised and the

800 absolute dryness channels, and the removal of the coordinate-informed channels. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main text here.

Coarsening the blocks from 1 km to 5 km moves the verdict counts from ten positive, seven negative and three uncertain to six, four and ten. Support
805 is removed from seven verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-
810 screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. This closes the one processing-artefact candidate for that region’s behaviour. Appendix A reports all seven arms, including one that tests a claim of Section 1.2 and does not uphold it.

815 4.8. *The same geography, a second fire: reversal with place held constant*

Every result above compares different places. Muğla admits a stricter test, because a second fire burned inside the identical AOI on the identical analysis grid. The 2021 event has a predictor window closing on 28 July. The 2022 event has a matched window closing on 20 June. Region, bounding box, cell
820 definition, feature registry and processing chain are the same in both arms. The design holds place fixed. It does not hold season fixed, because the 2022 fire ignites about six weeks earlier, and it does not hold the population fixed, because the 2022 arm is defined by removing the 2021 scar.

Elevation reverses with bootstrap support. In 2021 higher ground

Table 10: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 3’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 3 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

825 burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022
lower ground did, at 0.296 with its interval entirely below. The two inter-
vals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414,$
 $-0.220]$. The same predictor, the same region, the same grid, and an asso-
ciation that points the opposite way in two fires eleven months apart. The
830 four absolute thermal channels change side as well, from bootstrap-supported
lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because
they have no analogue in the twenty-direction matrix. The two arms are
not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and
835 elevation, slope and land cover are identical to the digit across all 73,098
grid cells, so only NDVI and the six thermal channels carry new information
between them. And the removal is the target’s own positive class, so in the
2022 to 2021 direction not one target positive is present in the source training
population while 38,789 of 38,819 target negatives are, and membership of
840 the source training set alone separates the 2021 target’s classes at ROC-
AUC 0.9996. We therefore cannot bound what that asymmetry does to the
two transfer numbers, only state that a direction with this structure is not
comparable to the twenty between-region directions. The elevation reversal is
the finding; the structural asymmetry is a competing explanation this design
cannot exclude. The known part of the bias runs the safe way: the removed
845 cells are high, with a median of 563 m, and unburned in 2022, so removing
them raises the 2022 elevation AUC and makes the reversal smaller rather
than larger.

4.9. *What target labels cost: the recovery curve*

850 Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many, for three regions across all six ordered directions, using one 10-cell (~ 5 km) 855 spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the 860 sixth. That budget is 7 to 20 % of the target's natural-vegetation population, about 800 km², so it is a real answer and not a cheap one. Two properties of the measurement bound it and are stated in the supplement rather than buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and 865 the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns. The protocol, the full curve and the limits are in Supplement S1.

Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels. 870 And six directions in three regions cannot support a general label budget, so none is offered.

5. Discussion

5.1. Principal findings

Three results carry this paper. First, the pre-fire thermal block raises
875 spatially blocked within-region ROC-AUC by +0.056 to +0.153 in every one
of five Mediterranean regions, with bootstrap support at 1 km and 5 km
blocking, and contributes +0.004 $[-0.028, +0.036]$ across twenty ordered
transfer directions, indistinguishable from zero and with a sign that varies
by pair. The static baseline transfers no better, at a mean of 0.537 against
880 0.541, so the failure is not a property of the dynamic block specifically. Nor
is it attributable to region crossing. On identical cells, withholding the burn
scar from training costs +0.082 $[+0.005, +0.159]$ and moving the training
data 306 to 2,802 km away costs $-0.003 [-0.063, +0.056]$. The failure is
a property of contiguous spatial holdout. The six below-chance directions
885 are a separate matter, since no account of merely lost skill produces a re-
liably reversed ranking. Removing the two reversing predictors, elevation
and the LST anomaly, costs -0.081 of within-region skill, supported in every
region and three quarters attributable to elevation, and changes transfer by
+0.014 $[-0.017, +0.045]$, which also spans zero. Second, of twenty candidate
890 transferability diagnostics only two have intervals excluding zero and both
are conditional, while no marginal measure was shown to order the matrix.
The marginal family, which includes area-of-applicability dissimilarity, is the
only one runnable before deployment. Third, the mechanism is a reversal in
the sign of the predictor-burning association, and it persists inside one study
895 area across two fires.

5.2. *Why the thermal increment is real but local*

A distinction has to be drawn before this section can say anything useful. The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows that most of the difference between them
900 is already present inside a single region. What follows therefore reads the increment as local in a specific sense: it holds where the held-out cells are interleaved with training cells, and most of it is gone once they are not, before the fire or the region changes. What that does not explain is why the directions differ from each other, and in particular why six of them are
905 anti-predictive.

The within-region increment is not an artefact to be explained away. It replicates in five independent regions, survives coarsening of the spatial blocks to ~ 5 km with its interval intact, persists in the secondary all-valid population, and survives a predictor window closed 7 and 14 days earlier in
910 all five regions. The direction of that last result is region-specific and should not be overstated: the contribution strengthens in Bejís ($0.058 \rightarrow 0.079$ at 14 days) and Muğla ($0.115 \rightarrow 0.128$), is flat in Montiferru, and **weakens monotonically in Evia** ($0.156 \rightarrow 0.149 \rightarrow 0.135$). What holds everywhere is survival, not improvement. The transfer failure is therefore not evidence
915 that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply different systems, so that a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems.
920 The two Muğla events answer it directly: region, AOI, grid, feature registry

and processing chain identical, static predictors identical cell by cell, and elevation’s association with burning reversing from 0.611 [0.532, 0.690] to 0.296 [0.230, 0.355], a difference of -0.317 [-0.414 , -0.220] (Section 4.8). Holding geography fixed does not stabilise the direction of the relationship.

925 Three caveats attach and none is dismissed. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells against a gate minimum of thirty. The population is
 930 not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And the 2022 arm rests on one compact scar and eleven positive-carrying 5 km blocks. What survives all three is the narrower reading that *something* reverses the elevation-burning association with geography, grid and pipeline held fixed, which is why Section 4.8 is reported as corroboration
 935 rather than as the load-bearing evidence for the mechanism.

The sharpest supported reversal belongs to elevation, a static predictor, so instability is a property of the predictor-to-burning mapping generally rather than of thermal channels specifically. This is the same conclusion the baseline transfer arm reaches from the other direction: the instability is not
 940 the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the question that matters when the failure is conditional. A target region can
 945 sit well inside the training envelope while the relationship between those

predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability and transfer below chance.

The same holds for the niche-overlap and regime families. At the point
950 estimates, the pair with the highest burned-niche overlap in the matrix
fails in both directions while the pair with the lowest transfers in both, so
no monotone function of overlap can order the outcomes. The qualifier is
not decorative: at the 5 km blocking this paper otherwise defends, neither
Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its
955 below-chance support, so this contrast is a statement about point estimates
and is made as one. The domain classifier is at ceiling for every pair, which
is informative in itself: the regions are trivially separable in predictor space
whether or not their transfer works, so separability carries no ordering infor-
mation here.

960 *5.4. What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against ob-
served transfer. Three limits are reported with it rather than after it. Its
tie structure caps the achievable Spearman at +0.861, so the observed value
sits essentially on its own ceiling and the ranking it supports is coarse. Its
965 exact one-sided permutation p is 0.0060, the smallest that tie structure can
produce, against a Bonferroni threshold of 0.0026 over nineteen computed
variants. No outcome could have cleared family-wise correction on ten ef-
fective pairs. And it needs burned labels in both regions, which makes it a
mechanism diagnosis rather than a pre-deployment screen.

970 That last limit is the substantive one, and it is why the paper's practical

conclusion is not "use this index instead". It is that the family of diagnostics that can be run before deployment is the family that fails here, and the information that would order transfer is the information a practitioner does not have when the decision is made. The honest statement is that these
975 twenty candidates were **not shown to order transfer**, not that they are shown incapable of it.

5.5. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below
980 the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and changes mean transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one.
985 The thermal block's own contribution to transfer is $+0.004$, with an interval spanning zero. Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Both arms are null on the portability axis. What the interventions measure is a local cost and no compensating transfer gain. That is not a conservation law, and it is not a rate at which local skill can
990 be sold for portability. No such rate is estimated here.

5.6. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null. The regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the

995 most regime-different pair transfers above chance. One mundane explanation
can be set aside: cutting Muğla, by far the largest population, to Manav-
gat’s cell count and separately to its positive count leaves Muğla’s transfer
behaviour inside the subsampling range in both roles. The error was in the
hypothesised grouping, not in the data, and with ten pairs this cannot refute
1000 regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [7] transfer successfully across a comparable Mediterranean
design and we do not, and the two results are not in conflict. Their predictors
are attributes of a place, ours describe the state of a surface in one season,
1005 which suggests that the relationship between domain similarity and transfer
success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here.
Their response variable is human-driven **ignition**; ours is burned **area**. Ig-
nition likelihood is dominated by human access and activity, stable and
1010 near-universal drivers, whereas burned extent is dominated by spread. A
predictor-class explanation and a response-variable explanation are there-
fore confounded in this comparison. Our own data speak against a simple
predictor-class reading in any case: the static baseline transfers at a mean
of 0.537 here, so within this cohort the place-attribute class does not travel
1015 either.

5.8. *Implications*

For practice, transfer skill has to be *measured* rather than inferred from
similarity, and the pre-deployment diagnostics currently used for that infer-

ence did not order it here. Where a model must be moved, the resource that
1020 closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to
89 % of the target’s own ceiling in three of six directions, and 30 to 57 %
in the rest. That is a real answer and not a cheap one, being 7 to 20 % of
the target’s natural-vegetation population, and at small budgets the same
intervention damages the direction that already transfers best.

1025 For method development, the results point away from better unsuper-
vised alignment. Two label-free methods applied carefully moved fourteen of
twenty directions towards chance, and every direction they improved involves
the smallest region. A sign reversal is not a distribution mismatch, and no
realignment of inputs can repair it.

1030 5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the
boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say
how local skill and portability behave for a mixed thermal-plus-weather pre-
1035 dictor set.

(ii) **Temporal transfer is measured for one region only**, and even
there year and seasonal phase are confounded, a confound that cannot be
resolved in this study area for the reason given in Section 5.2. Its 331 burned
cells also leave the thermal reversals unresolved at interval level, and the
1040 pair holds place fixed but not population. Those two arms are addition-
ally the only transfer directions here computed by us rather than read from
the pipeline author’s frozen export, with his unmodified code and the same
pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
1045 [9], whose omission and commission characteristics [3] bound every model
evaluated here. No second burned-area product covers 2021 and 2022 at this
resolution; the companion paper reports what an independent active-fire ob-
servation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even
1050 after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072
in three of the others.

(v) **Each region contributes one fire season**, so regional concept
shift is confounded with event meteorology, and distinguishing them requires
multi-year labels.

1055 (vi) **Cross-region point estimates carry an implementation tol-
erance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported
numbers are fixed to one verified version, but exact reproduction elsewhere
requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of
1060 ten region pairs.** Both the successes and the failures of Section 4.4 should
be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.**
It is where the conditional diagnosis bites hardest and where feature removal
recovers most. Two candidates have now been tested and neither survives.
1065 The first was its meteorology: the predictor window was not extreme. The
second was the quality-screening difference in its coarse thermal input. It was
the more worrying of the two, because the change it induces correlates with
elevation at +0.615. It does not propagate. The region’s entire downstream

chain was rebuilt from a quality-screened MODIS input. That changes the
1070 downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed
univariate association moves by more than +0.0003. Elevation’s is identical
in both arms, at 0.374, and the population is unchanged. The result is
structural rather than fortunate. Elevation is a DEM variable the screening
cannot touch, and fusion falls back on the MODIS-derived surface across
1075 only 2.14 percentage points of coverage. With one fire season per region the
remaining candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point
estimates behind them.** Several verdicts sit within a thousandth of their
reference value, and at 1 km blocking the published split of ten positive,
1080 seven negative and three uncertain turns on a lower bound of -0.00045 . The
point estimates and the sign pattern are stable; the counts are not. Every
sentence in this paper that leans on an exact count of supported directions
should be read at that precision.

6. Conclusions

1085 Pre-fire thermal dryness adds a real and repeatable increment to burned-
area discrimination. Across five Mediterranean regions, ROC-AUC was raised
by +0.06 to +0.15 over a static and near-static terrain, fuel and greenness
baseline. That baseline’s only time-varying member is the vegetation-index
composite. The gain survived spatial blocking at about 5 km, the coarsest
1090 scale this design supports as an interval. It also survived a predictor window
closed up to two weeks before the first labelled burning.

That skill is local. Paired per direction, the same predictor block adds

nothing distinguishable from zero to cross-region transfer. The mean is
+0.004 over twenty directions and its interval spans zero under all four
1095 between-direction resampling units we computed. Its contribution changes
sign from one direction to another. The paired deltas run from -0.148 to
+0.132, twelve positive and eight negative. At the conservative 5 km block-
ing, five to six directions are helped and three to four are harmed with interval
support, with the rest carrying no verdict. The block is also the swing factor
1100 at the chance line.

Two controls fix the meaning of all this. The baseline arm transfers at a
mean of 0.537 against the thermal model's 0.541, so the static predictor class
is not portable here either. And three evaluations on identical cells locate
where it goes: 0.627 with the held-out burn scar in the training data, 0.552
1105 with it withheld, and 0.559 for a model fitted 306 to 2,802 km away. The
honest summary is therefore not that these predictors fail to cross regions.
The failure is a property of contiguous spatial holdout: the fire-
specific residual is $+0.082$ [$+0.005$, $+0.159$] and the 2,800 km costs -0.003
[-0.063 , $+0.056$]. Two thirds of the apparent collapse from a region-wide
1110 0.782 is the evaluation area itself, which is 34 to 87 per cent burned against
3.8 to 28.7 for a region. Six directions are nonetheless anti-predictive with
interval support, which no account of merely lost skill explains. The one-
event-per-region design also means that a fire cannot be separated here from
the season and meteorology that produced it.

1115 Removing the two reversing predictors, elevation and the LST anomaly,
costs -0.081 of mean within-region skill, with interval support in every region
and roughly three quarters of it attributable to elevation, a baseline terrain

variable; it changes mean transfer by +0.014, an estimate whose interval spans zero. A local cost is measured. No compensating transfer gain is, on
1120 either arm, so no exchange between the two is demonstrated.

The relationship itself is also unstable, and that is a separate finding. The direction of the link between dryness and burning changes from one region to another, with bootstrap support for two predictors. This is measured on univariate associations and does not depend on any fitted model, so the
1125 distance result above neither establishes it nor removes it. What the two together rule out is the comfortable reading in which regional concept shift explains the whole transfer matrix. Most of its *level* is reached without leaving a region. What varies around that level, including six anti-predictive directions, is not explained by separation and is where the instability matters.

1130 None of the similarity diagnostics tested here ordered the transfer matrix: predictor-space distance, domain separability, niche overlap and regime structure all failed. On ten effective pairs those null results mean not shown to order transfer, rather than shown not to. Label-blind adaptation compresses transfer towards chance instead of repairing it, and pooled multi-region training
1135 does not escape it.

The practical implication is a change in what is checked before a dynamic-state fire model is transferred. The question is not whether the target region lies inside the source’s environmental envelope. In our matrix, the pair with the highest envelope overlap failed in both directions, and the pair with the
1140 lowest overlap transferred in both. Both statements are made at the point estimate. The question is whether the signed feature-response directions agree.

That check requires a labelled probe in the target region, and so, on inspection, do the niche and regime diagnostics. Only the marginal family
1145 can be computed before any target label exists, and it is the family that failed. How large the probe has to be is not established here, so no label budget is claimed. A signed association with a usable interval may need fewer labels than a refitted model, but that was not tested. What is clear is that the supervised recalibration that label-free alignment cannot deliver is
1150 not cheap. Section 4.9 prices it for three regions: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target ceiling in three of six directions, 51 to 57 % in two more and 30 % in the sixth, and at its top budget the labelled set already holds most of one target region’s burned cells. Where no target labels exist, the transfer performance of such models should be treated as
1155 unknown rather than inferred from similarity.

For model builders, local skill and portability should be reported separately rather than assumed to travel together. A predictor block worth a large within-region increment may contribute nothing distinguishable from zero across regions, with a sign that varies by pair, and reporting only within-
1160 region validation hides that completely. Three extensions are left for future work: temporal transfer, meteorological covariates, and physically normalised dryness variables. Each is a step towards the self-calibrating thermal monitoring system that motivated this study.

Appendix A. Appendix A. Sensitivity analyses

1165 Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a

reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a claim the introduction makes, and the claim is not upheld.

1170 **(a) Evia AOI and prevalence.** The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

1175 **(b) CORAL regularisation, including the canonical $\lambda = 1$.** Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ of the cited method lies outside that sweep and was computed separately on the same four directions. It changes the thermal
1180 mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of -0.005 to $+0.016$, and **no direction crosses the chance line**. The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at $\lambda = 0.1$. No CORAL-dependent conclusion in this paper is sensitive to λ over the full range from 0 to 1.

1185 **(c) Blocking scale.** Recomputing the transfer quantities at 10-cell (≈ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes support from seven verdicts and adds none. Across five seeds the counts run
1190 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a

result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The
1195 comparison does establish two things. Every verdict that changes, changes towards "no verdict". And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this variation.

1200 **(d) Predictor-window closure.** The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in
1205 Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-
1210 correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation is identical in both arms, at 0.374, the population is unchanged, and the
1215 within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$. The reason is structural. Elevation is a DEM variable the screening cannot

touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

1220 **(f) Normalised against absolute dryness channels.** Section 1.2 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as a feature-set contrast.

1225 One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Section 4.8, where the absolute channels move. That is a statement about one region across two fires. **Across regions it does not hold:** `1st_`
1230 `anomaly_mean` is itself one of the two bootstrap-supported reversing predictors, reversing between Bejís and Evia, which is why Section 4.6b drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

1235 Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

1240 **The advantage is not found.** Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and

Table A.11:

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.3: its

mean transfer of 0.5371 was recomputed here from the modelling datasets
1255 and matches the 0.537 read from the frozen per-direction export. Details are
in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one
cohort. It does not test normalised dryness indices in general, and it does
not test a normalisation fitted against a pooled multi-region reference rather
1260 than each region's own baseline years.

(g) The coordinate-informed channels. `downscaled_lst_mean` and
`fused_lst_mean` come from a per-region downscaling model whose own in-
puts include coordinates, which Section 3.14 names as the one route by which
a coordinate-derived surface re-enters a feature set that excludes coordinates.
1265 A coordinate-smoothed surface is by construction locally informative and
non-portable, so if the within-region increment depended on it, that incre-
ment would be this paper's own result in miniature rather than a finding
about thermal dryness. The arm was therefore run here rather than de-
ferred.

1270 Both channels were dropped and everything refitted, over all twenty direc-
tions and all five within-region folds. The reference configuration reproduces
the frozen exports exactly, at 0.000000 against both the transfer and the
within-region references.

The increment does not depend on them. It stays positive in every
1275 region and retains 82 % to 103 % of its size, rising slightly in one region. Mean
transfer is likewise unchanged, at 0.545 without the two channels against
0.541 with them and 0.537 for the baseline alone. The local skill this paper
reports is therefore not an artefact of a coordinate-smoothed surface.

Table A.12:

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

This arm was computed independently here and reproduces the companion paper’s figure for the same quantity, 82 % to 103 %, from a separately written harness.

Appendix A.1. A(h). The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models as a ladder. The per-split and per-scar detail is here, so that the ladder can be checked without leaving the manuscript.

Table A1. Within-region half-split, every split. Source and target positive counts are given because they are unequal, which is the principal limit on this arm: a straight cut does not produce two exchangeable halves. Two splits are unusable because one half of Manavgat contains no burned cells.

Table A2. Leave-one-scar-out at a 2 km buffer, every scar. Only Muğla contains more than one burned component of at least 50 cells, so it is the only region where holding one out leaves the source model properly

Table A.13:

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

trained. Its four arms mean 0.579; the four arms in the other regions mean
1295 0.525, and the pooled figure of 0.552 averages the two.

Table A.14:

Region	Component	Source	Target positives	Target cells	AUC
		positives			
		left			
evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of the two arms are not comparable with each other or with the transfer targets. A held-out scar with its 2 km collar runs at 43 to 87 % burned prevalence; a whole target region runs at 3.8 to 28.7 %. The scar arm therefore asks for discrimination against the nearest and most
1300 similar negatives only, while a transfer arm includes the whole easy far field. Section 4.3 states the consequence: the last three rows of the ladder are not distinguishable by this design.

Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.4 summarises this table by family. The full ranking is given
1305 here because the negative result is the point: a reader should be able to see

every candidate that was tried, not only the families' best members, and to check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

Reading the two rows that clear zero. Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.4 says so.

Equal-sample check. The families sit on unequal samples: marginal, applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5}

Table B.15:

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	$P(y x)$ conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	$P(y x)$ conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	$P(y x)$ conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector	$P(y x)$ conditional	+	20	+0.27 [−0.36, +0.77]	no
Spearman, all 9					
Agreement count, all 9	$P(y x)$ conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	$P(x y=1)$ niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	$P(x y=1)$ niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	$P(x y=1)$ niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	$P(x y=1)$ niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	$P(x y=1)$ niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	$P(x)$ marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	$P(x)$ marginal	−	12	−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	$P(x)$ marginal	−	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	$P(x)$ marginal	+	12	+0.22 [−0.48, +0.59]	no
Fraction inside	$P(x)$ marginal	+	12	+0.08 [−0.89, +0.43]	no

and leaves the ordering unchanged — the conditional rows still lead at +0.87
 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero.
 Source: `paper/diagnostics_common_subset.md`.

*Appendix B.1. B2. The signed univariate associations the reversal claim
 rests on*

Section 4.6b drops two predictors because their signed association with
 burning reverses between regions with bootstrap support, and Contribution
 3 names the reversal as the mechanism. This is the evidence for both. It was
 computed for the frozen analysis and is reproduced here because the claim
 is otherwise asserted rather than shown.

**Table B2. Signed univariate AUC of each predictor against
 burned, by region.** Primary natural-vegetation population; 10-cell (~5
 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never
 folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values
 rank burned and is a direction rather than weakness. **Bold** marks a region
 whose own interval excludes 0.5.

What counts as a reversal. A pair of regions is called a reversal
 only when their signed associations point to opposite sides of 0.5 **and each
 region’s own interval excludes 0.5**. That is stricter than requiring the
 two regions’ intervals to be disjoint, and the difference matters: for `current_
 1st_mean` between Manavgat and Muğla the two intervals are disjoint, but
 Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no
 established direction to reverse from. That pair is a point reversal, not a
 supported one.

Table B.16:

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_	0.374 [0.289,	0.643 [0.558,	0.611 [0.532,	0.541 [0.448,	0.584 [0.395,
mean	0.471]	0.729]	0.690]	0.626]	0.762]
slope_mean	0.531 [0.423,	0.521 [0.439,	0.637 [0.582,	0.487 [0.418,	0.652 [0.506,
	0.642]	0.605]	0.686]	0.554]	0.771]
ndvi_mean	0.636 [0.587,	0.559 [0.497,	0.662 [0.616,	0.639 [0.575,	0.586 [0.450,
	0.676]	0.619]	0.704]	0.701]	0.704]
lst_anomaly_	0.482 [0.428,	0.418 [0.364,	0.485 [0.395,	0.640 [0.567,	0.395 [0.285,
mean	0.530]	0.480]	0.566]	0.710]	0.535]
current_lst_	0.538 [0.452,	0.477 [0.401,	0.325 [0.271,	0.377 [0.301,	0.370 [0.248,
mean	0.621]	0.547]	0.382]	0.456]	0.513]
current_	0.552 [0.460,	0.517 [0.429,	0.336 [0.275,	0.362 [0.285,	0.356 [0.233,
tvdi_mean	0.641]	0.595]	0.398]	0.442]	0.499]
tvdi_	0.449 [0.391,	0.512 [0.443,	0.490 [0.396,	0.519 [0.444,	0.378 [0.282,
difference_	0.505]	0.583]	0.575]	0.589]	0.497]
mean					
downscaled_	0.552 [0.466,	0.484 [0.400,	0.307 [0.253,	0.377 [0.297,	0.365 [0.240,
lst_mean	0.637]	0.560]	0.366]	0.459]	0.511]
fused_lst_	0.540 [0.454,	0.481 [0.404,	0.325 [0.272,	0.376 [0.300,	0.370 [0.248,
mean	0.622]	0.551]	0.383]	0.456]	0.513]

1355 **Table B3. The cross-region reversals that meet the stricter criterion.** Difference intervals are from the same paired bootstrap.

Table B.17:

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not
1360 counted. The conservative criterion costs the paper findings rather than manufacturing them: a difference interval on the pair, which is the instrument Section 4.8 uses, would support more reversals than the three listed here.

Appendix C. Appendix C. Protocol detail

These protocols are given here in full rather than in Methods, because
1365 each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

1370 Labels come from MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`)
retrieved through Google Earth Engine [10], whose omission and commission
characteristics [3] bound every model here. The product is exported onto
the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which
duplicates each native ~ 500 m observation across the 30 m pixels beneath
1375 it.

The analysis grid is therefore **reconstructed rather than native**: the
30 m grid is partitioned into non-overlapping blocks of `round(500 / 30) =`
`17 × 17` pixels, giving a nominal cell edge of 510 m, and each cell is identified
by its integer block indices `row_500m` and `col_500m`. This approximates the
1380 MODIS sinusoidal cell rather than reproducing it, being anchored to the
Landsat/EPSC:4326 reference grid, and marginal blocks are truncated. The
cell is square in degrees but not on the ground, about 510 m north to south
and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a
MODIS cell’s 0.250 km²; every block-size label in this paper is therefore the
1385 north-south dimension. The companion paper develops what follows from
that.

A cell’s representative burn date is the **mode** of the positive sub-pixel
day-of-year values in the block, tested against the region’s label window. Be-
cause the exported raster carries no out-of-window positives, in this dataset
1390 a single in-window positive sub-pixel makes the cell burned. The fraction of
positive sub-pixels agreeing with the modal date is recorded as `burn_date_`
`pixel_agreement_fraction` but no agreement threshold is imposed. The

label never affects a cell’s eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

1395 Two safeguards are recorded rather than assumed. Cells that burned inside a region’s own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort; 1400 the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction 1405 reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which 1410 MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from four families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The 1415 families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [21, 22, 16], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [34] and Warren’s I [41] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening

on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented
1445 **as one.** The 2022 event ignites about six weeks earlier in the season than
the 2021 event, so calendar year and seasonal phase are confounded and no
difference can be attributed to the year alone. The registered claim is same-
geography event-to-event. The pair is deliberately kept out of the twenty-
direction matrix, and that exclusion is enforced in released code rather than
1450 merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion
applies to the 2022 arm alone, masking every cell that burned in 2021 so the
previous year’s scar does not enter the following year’s analysis: it removes
3,073 cells, of which 2,941 belong to the primary population, taking it from
1455 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of
38,790 cells and three byte-identical static predictors, and in the 2022-to-2021
direction every target positive lies outside the source training population.
Section 4.8 states what follows for how these two numbers may be read.

CRedit authorship contribution statement

1460 **Yunus Emre Cogurcu:** Conceptualization, Methodology, Formal anal-
ysis, Investigation, Writing – original draft, Writing – review and editing,
Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
idation, Writing – review and editing.

Declaration of competing interest

1465 The authors declare that they have no known competing financial inter-
ests or personal relationships that could have appeared to influence the work
reported in this paper.

Funding

This work is an output of project 17506, supported by the Scientific Re-
1470 search Projects (BAP) unit of Çukurova University.

Data availability

All satellite inputs are public and are obtained through Google Earth
Engine. The processing pipeline, its configuration and the frozen numeric
outputs are publicly available; the repository, the commit of record and the
1475 licence are given in the data and code availability statement of the Methods,
together with the three components that fall outside that release. No digital
object identifier is minted and no archival deposit exists.

References

- [1] Alkan Akıncı, H., Akıncı, H., 2023. Machine learning based forest fire
1480 susceptibility assessment of Manavgat district (Antalya), Turkey. *Earth
Science Informatics* 16, 397–414. doi:10.1007/s12145-023-00953-5.
- [2] Archibald, S., Lehmann, C.E.R., Gómez-Dans, J.L., Bradstock, R.A.,
2013. Defining pyromes and global syndromes of fire regimes. *Proceed-
ings of the National Academy of Sciences* 110, 6442–6447. doi:10.1073/
1485 pnas.1211466110.

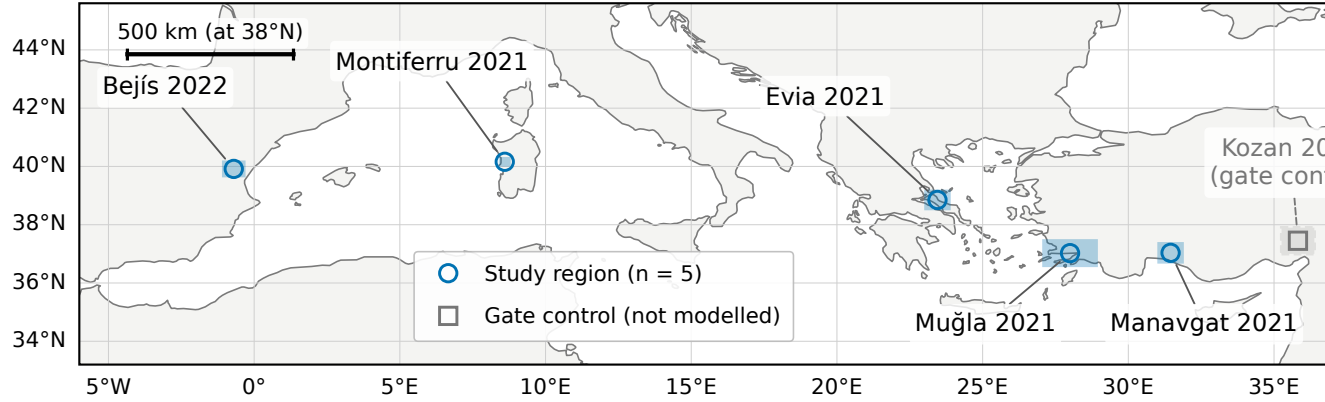


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

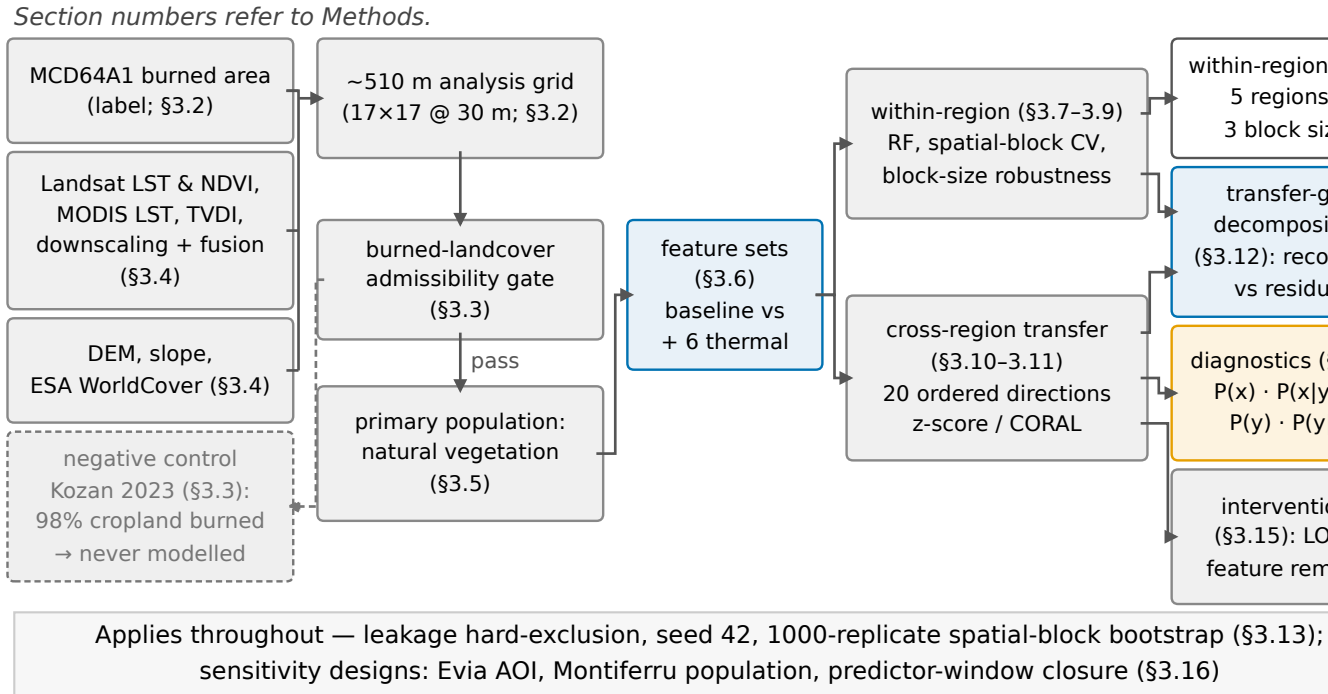


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

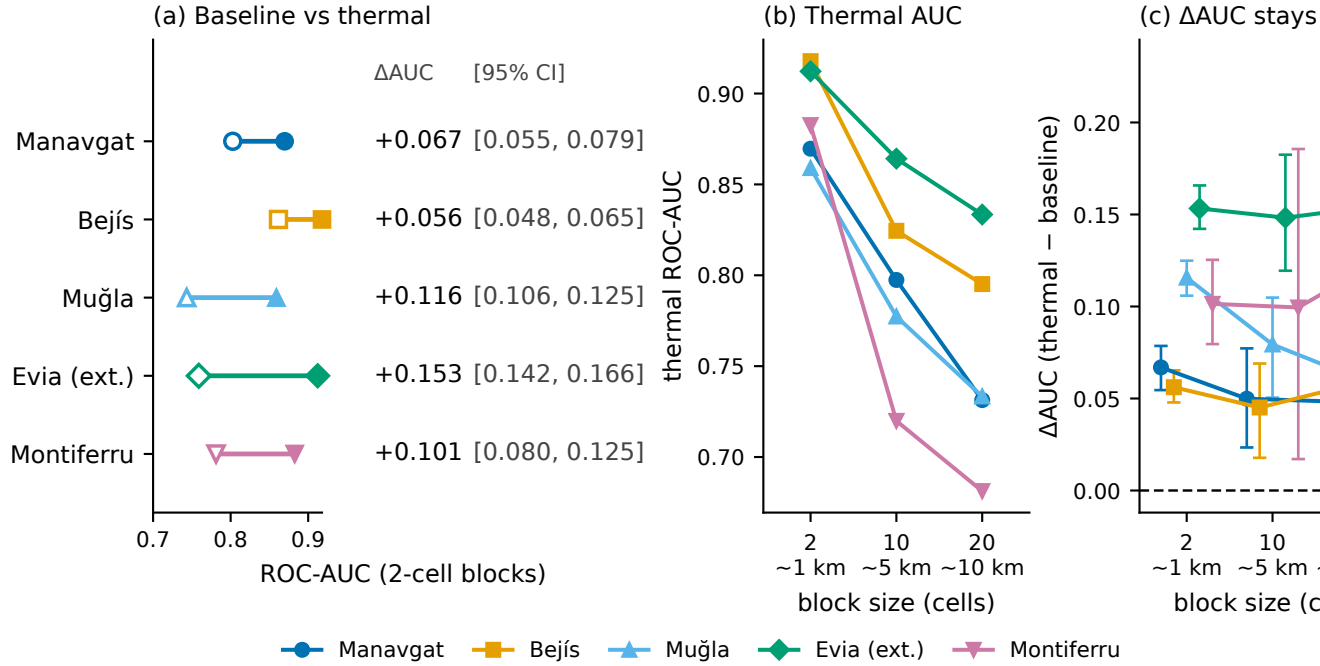


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx 1, 5$ and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 3, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

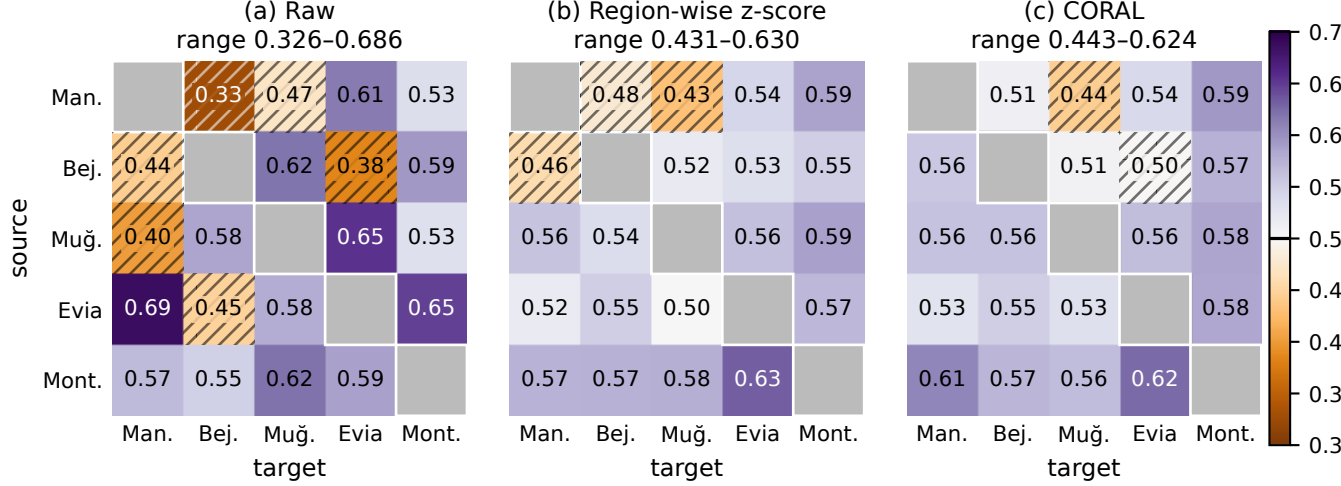


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

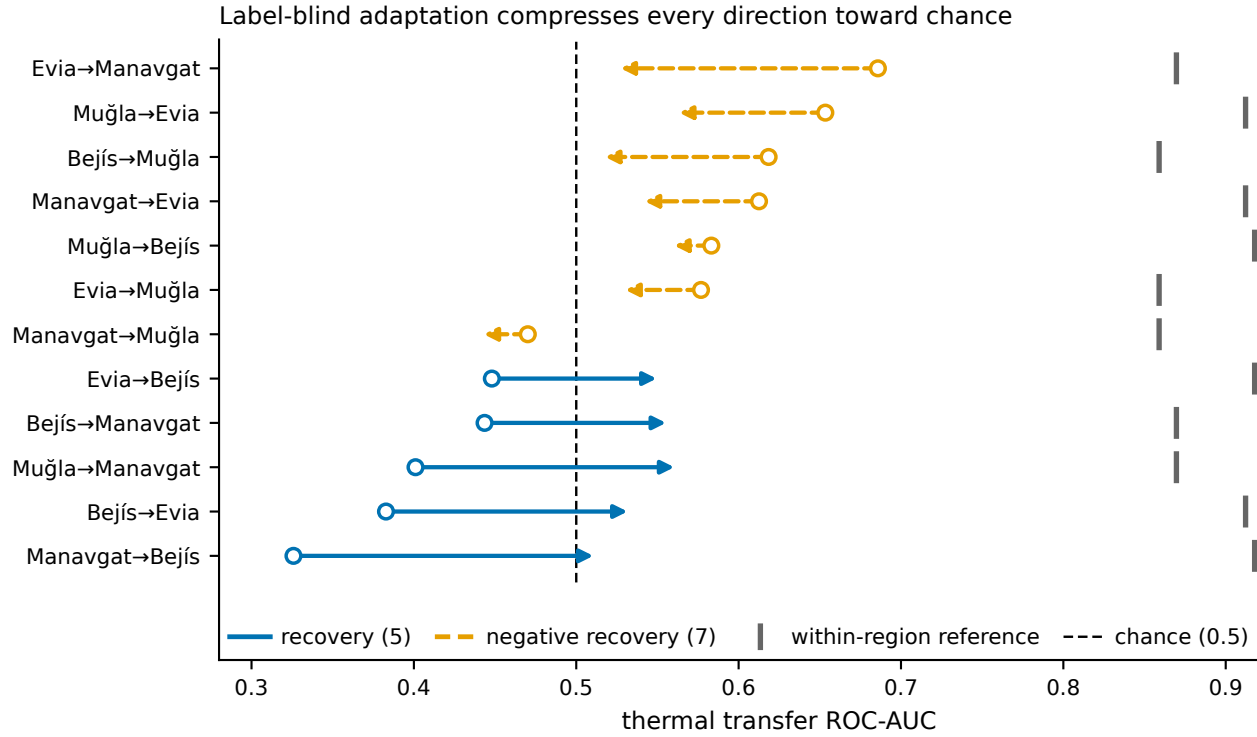


Figure C.5: **Label-blind adaptation compresses transfer toward chance rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

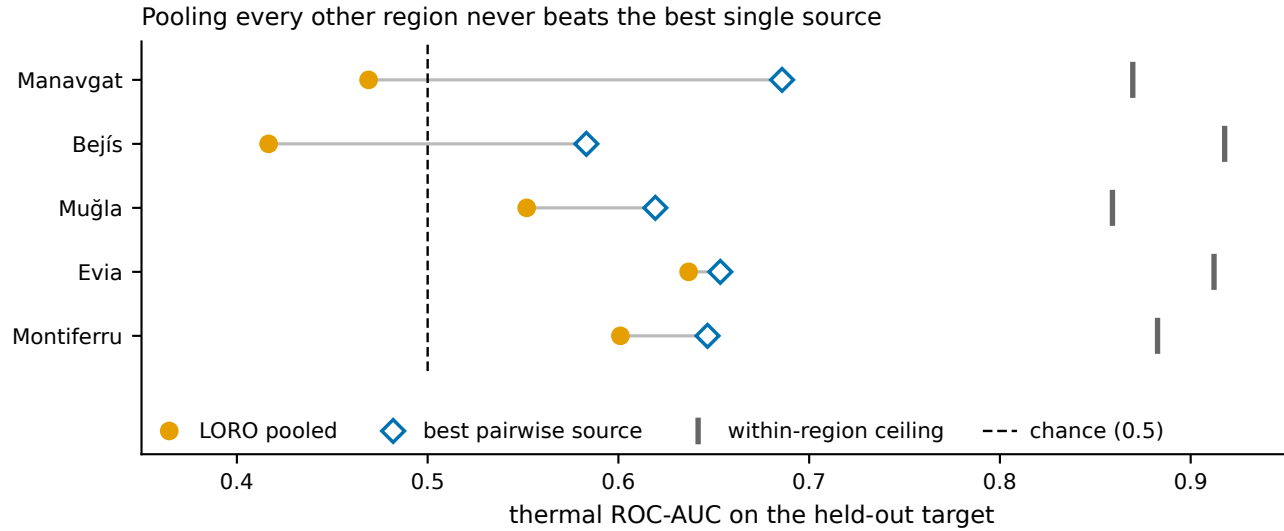


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

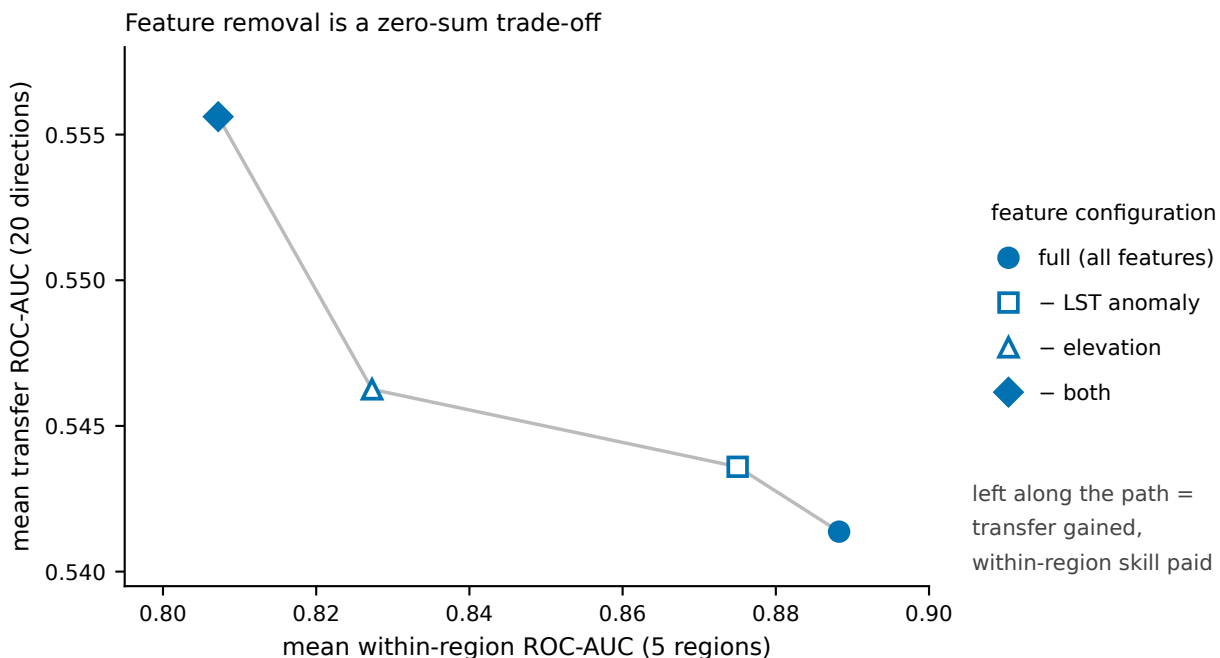


Figure C.7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

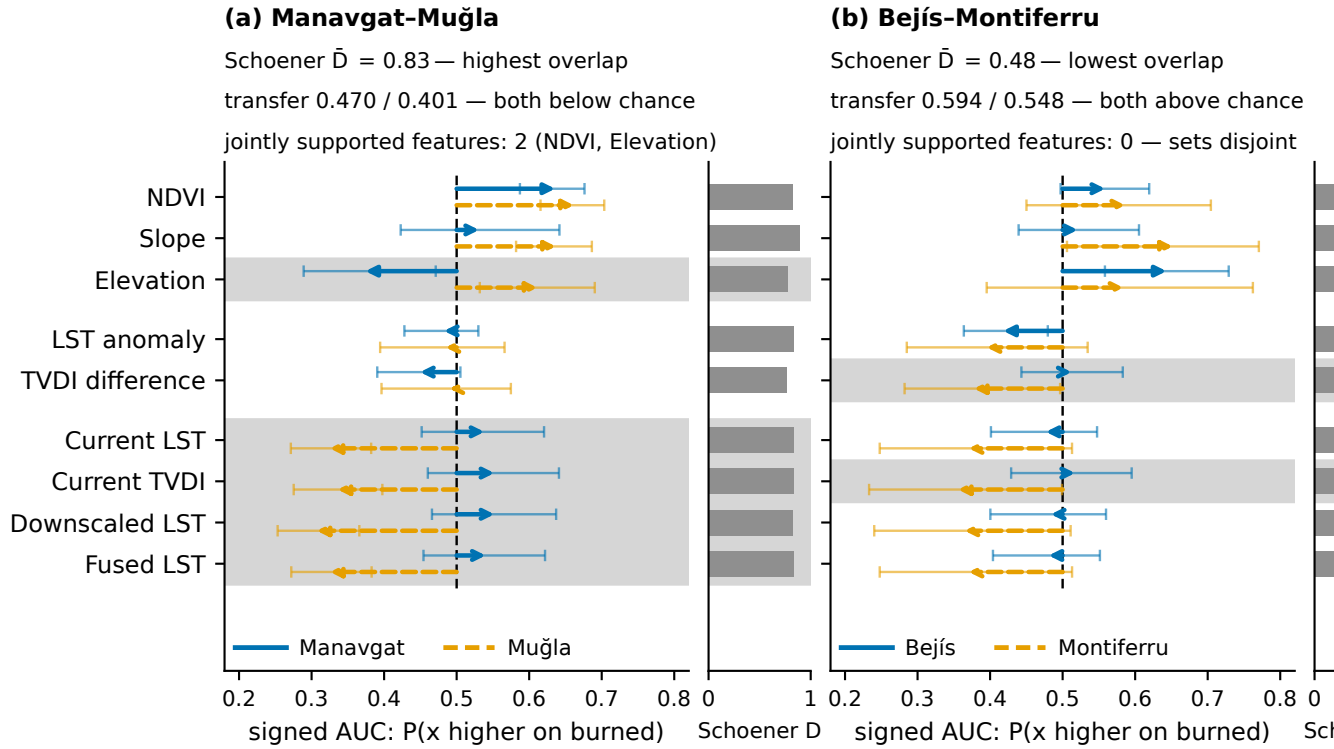


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

- [3] Boschetti, L., Roy, D.P., Giglio, L., Huang, H., Zubkova, M., Humber, M.L., 2019. Global validation of the Collection 6 MODIS burned area product. *Remote Sensing of Environment* 235, 111490. doi:10.1016/j.rse.2019.111490.
- 1490 [4] Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:1010933404324.
- [5] de Bruin, S., Brus, D.J., Heuvelink, G.B.M., van Ebbenhorst Tengbergen, T., Wadoux, A.M.J.C., 2022. Dealing with clustered samples for assessing map accuracy by cross-validation. *Ecological Informatics* 69, 101665. doi:10.1016/j.ecoinf.2022.101665.
- 1495 [6] Chuvieco, E., Cocero, D., Riaño, D., Martín, P., Martínez-Vega, J., de la Riva, J., Pérez, F., 2004. Combining NDVI and surface temperature for the estimation of live fuel moisture content in forest fire danger rating. *Remote Sensing of Environment* 92, 322–331. doi:10.1016/j.rse.2004.01.019.
- 1500 [7] Dimarco, N.A., Faraji, I., Wahbi, M., Maatouk, M., Boulaassal, H., Yazidi Aalaoui, O., El Kharki, O., 2026. Generalizing human-driven wildfire ignition models across Mediterranean regions using harmonized remote-sensing and machine-learning data. *Geomatics* 6, 13. doi:10.3390/geomatics6010013.
- 1505 [8] Gelabert, P.J., Jiménez-Ruano, A., Ochoa, C., Alcasena, F., Sjöström, J., Marrs, C., Ribeiro, L.M., Palaiologou, P., Bentu Martínez, C., Chuvieco, E., Vega-García, C., Rodrigues, M., 2025. Assessing

- human-caused wildfire ignition likelihood across Europe. *Natural Hazards and Earth System Sciences* 25, 4713–4729. doi:10.5194/nhess-25-4713-2025.
- [9] Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. *Remote Sensing of Environment* 217, 72–85. doi:10.1016/j.rse.2018.08.005.
- [10] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. doi:10.1016/j.rse.2017.06.031.
- [11] Huang, Y., Liu, N., Chen, W., Tan, W., Deng, Y., Wang, Z., Chlus, A., Shen, J., Townsend, P.A., 2026. Enhancing transferability of foliar trait retrieval models: A comparative analysis of transfer learning strategies and domain shift characterization. *Remote Sensing of Environment* 337, 115336. doi:10.1016/j.rse.2026.115336.
- [12] Iban, M.C., Sekertekin, A., 2022. Machine learning based wildfire susceptibility mapping using remotely sensed fire data and GIS: A case study of Adana and Mersin provinces, Turkey. *Ecological Informatics* 69, 101647. doi:10.1016/j.ecoinf.2022.101647.
- [13] Jodhani, K., Bhatiya, L., Sirvi, L., Chanda, A., Thakkar, S., Rathnayake, U., 2026. Artificial intelligence and remote sensing frameworks for wildfire monitoring and risk analysis across multiple ecosys-

tems: A review. *AIMS Environmental Science* 13, 56–98. doi:10.3934/environsci.2026004.

[14] Kondylatos, S., Prapas, I., Camps-Valls, G., Papoutsis, I., 2023. Mesogeos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean, in: *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023) Datasets and Benchmarks Track, pp. 50661–50676. doi:10.52202/075280-2204.

[15] Liu, C., Mostafavi, A., 2025. WildfireGenome: Interpretable machine learning reveals local drivers of wildfire risk and their cross-county variation. *arXiv preprint arXiv:2511.11589*. doi:10.48550/arXiv.2511.11589, arXiv:2511.11589.

[16] Ludwig, M., Moreno-Martinez, A., Hölzel, N., Pebesma, E., Meyer, H., 2023. Assessing and improving the transferability of current global spatial prediction models. *Global Ecology and Biogeography* 32, 356–368. doi:10.1111/geb.13635.

[17] Maffei, C., Alfieri, S.M., Menenti, M., 2018. Relating spatiotemporal patterns of forest fires burned area and duration to diurnal land surface temperature anomalies. *Remote Sensing* 10, 1777. doi:10.3390/rs10111777.

[18] Maffei, C., Lindenbergh, R., Menenti, M., 2021. Combining multi-spectral and thermal remote sensing to predict forest fire characteristics. *ISPRS Journal of Photogrammetry and Remote Sensing* 181, 400–412. doi:10.1016/j.isprsjprs.2021.09.016.

- 1555 [19] Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and rate of spread from pre-fire multispectral satellite measurements. *ISPRS Journal of Photogrammetry and Remote Sensing* 158, 263–278. doi:10.1016/j.isprsjprs.2019.10.013.
- [20] Marino, E., Yáñez, L., Guijarro, M., Madrigal, J., Senra, F., Rodríguez, S., Tomé, J.L., 2024. Transferability of empirical models derived from
1560 satellite imagery for live fuel moisture content estimation and fire risk prediction. *Fire* 7, 276. doi:10.3390/fire7080276.
- [21] Meyer, H., Pebesma, E., 2021. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* 12, 1620–1633. doi:10.1111/2041-210X.13650.
1565
- [22] Meyer, H., Pebesma, E., 2022. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Communications* 13, 2208. doi:10.1038/s41467-022-29838-9.
- [23] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using
1570 forward feature selection and target-oriented validation. *Environmental Modelling & Software* 101, 1–9. doi:10.1016/j.envsoft.2017.12.001.
- [24] Milà, C., Mateu, J., Pebesma, E., Meyer, H., 2022. Nearest neighbour distance matching leave-one-out cross-validation for map validation.
1575 *Methods in Ecology and Evolution* 13, 1304–1316. doi:10.1111/2041-210X.13851.

- [25] Moreno-Torres, J.G., Raeder, T., Alaiz-Rodríguez, R., Chawla, N.V., Herrera, F., 2012. A unifying view on dataset shift in classification. *Pattern Recognition* 45, 521–530. doi:10.1016/j.patcog.2011.06.019.
- 1580 [26] Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 4349–4383. doi:10.5194/essd-13-4349-2021.
- 1585 [27] Pausas, J.G., Keeley, J.E., 2021. Wildfires and global change. *Frontiers in Ecology and the Environment* 19, 387–395. doi:10.1002/fee.2359.
- [28] Persello, C., Bruzzone, L., 2012. Active learning for domain adaptation in the supervised classification of remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing* 50, 4468–4483. doi:10.1109/TGRS.2012.2192740.
- 1590 [29] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., Pélissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications* 11, 4540. doi:10.1038/s41467-020-18321-y.
- 1595 [30] Podschwit, H., Jolly, W., Alvarado, E., Verma, S., Ponce, B., Markos, A., Aliaga-Nestares, V., Rodriguez-Zimmermann, D., 2022. Reliability

- of cross-regional applications of global fire danger models: A Peruvian
 1600 case study. *Fire Ecology* 18, 25. doi:10.1186/s42408-022-00150-7.
- [31] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-
 Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller,
 W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-
 validation strategies for data with temporal, spatial, hierarchical, or
 1605 phylogenetic structure. *Ecography* 40, 913–929. doi:10.1111/ecog.
 02881.
- [32] Rousseau, J.S., Betts, M.G., 2022. Factors influencing transferability
 in species distribution models. *Ecography* 2022, e06060. doi:10.1111/
 ecog.06060.
- 1610 [33] Sandholt, I., Rasmussen, K., Andersen, J., 2002. A simple interpreta-
 tion of the surface temperature/vegetation index space for assessment of
 surface moisture status. *Remote Sensing of Environment* 79, 213–224.
 doi:10.1016/S0034-4257(01)00274-7.
- [34] Schoener, T.W., 1968. The *Anolis* lizards of Bimini: Resource partition-
 1615 ing in a complex fauna. *Ecology* 49, 704–726. doi:10.2307/1935534.
- [35] Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy do-
 main adaptation, in: *Proceedings of the Thirtieth AAAI Conference on
 Artificial Intelligence (AAAI-16)*, AAAI Press, Phoenix, Arizona, USA.
 pp. 2058–2065. doi:10.1609/aaai.v30i1.10306.
- 1620 [36] Tuia, D., Persello, C., Bruzzone, L., 2016. Domain adaptation for the
 classification of remote sensing data: An overview of recent advances.

IEEE Geoscience and Remote Sensing Magazine 4, 41–57. doi:10.1109/MGRS.2016.2548504.

- [37] Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G., 2019. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution* 10, 225–232. doi:10.1111/2041-210X.13107.
- [38] Vesk, P.A., Morris, W.K., Neal, W.C., Mokany, K., Pollock, L.J., 2021. Transferability of trait-based species distribution models. *Ecography* 44, 134–147. doi:10.1111/ecog.05179.
- [39] Vibhandik, P., Sawant, S., Joshi, A., Bidwe, R., 2026. A systematic literature review on forest fire susceptibility mapping using geo-spatial technology and future research directions. *Discover Artificial Intelligence* 6, 921. doi:10.1007/s44163-026-00921-0.
- [40] Wadoux, A.M.J.C., Heuvelink, G.B.M., de Bruin, S., Brus, D.J., 2021. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling* 457, 109692. doi:10.1016/j.ecolmodel.2021.109692.
- [41] Warren, D.L., Glor, R.E., Turelli, M., 2008. Environmental niche equivalency versus conservatism: Quantitative approaches to niche evolution. *Evolution* 62, 2868–2883. doi:10.1111/j.1558-5646.2008.00482.x.
- [42] Xu, Y., Dai, Y., Chang, L., Wang, Q., Dong, Y., 2026. Does your wildfire prediction model actually work, or just score well?

- 1645 arXiv preprint arXiv:2605.18911. doi:10.48550/arXiv.2605.18911,
arXiv:2605.18911.
- [43] Yates, K.L., Bouchet, P.J., Caley, M.J., Mengersen, K., Randin, C.F.,
Parnell, S., et al., 2018. Outstanding challenges in the transferability of
ecological models. *Trends in Ecology & Evolution* 33, 790–802. doi:10.
1650 1016/j.tree.2018.08.001.
- [44] Yebra, M., Dennison, P.E., Chuvieco, E., Riaño, D., Zylstra, P., Hunt,
E.R., Danson, F.M., Qi, Y., Jurdao, S., 2013. A global review of remote
sensing of live fuel moisture content for fire danger assessment: Moving
towards operational products. *Remote Sensing of Environment* 136,
1655 455–468. doi:10.1016/j.rse.2013.05.029.
- [45] Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W.,
Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz,
S., Lesiv, M., Herold, M., Tsendbazar, N.E., Xu, P., Ramoino, F., Arino,
O., 2022. ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/
1660 zenodo.7254221.